The empirical recurrence for OEIS A282557: recovering a conjecture that is not written down

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Abstract

OEIS A282557 records "Empirical recurrence of order 62", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 142$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 62, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 62 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A282557 is "Number of nX5 0..1 arrays with no 1 equal to more than two of its king-move neighbors, with the exception of exactly one element.". It has offset 1 and begins

$$0, 154, 4167, 73140, 1381258, 23705784, 395048648, 6468244262, \ \ldots$$

The entry states:

> Empirical recurrence of order 62 (see link above)

As of the "Last modified" line on the live entry (Feb 18 2017 08:39:18, revision 4) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1132$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 142$ of the 1132, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 142$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 284$ exact terms of the model returns order 62 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{62} c_i \, a(n-i),$$

c_1	28	c_{17}	−10908620548	c_{33}	−33989588294784	c_{49}	594873111694
c_2	−174	c_{18}	38776306382	c_{34}	49308986790768	c_{50}	−306814807019
c_3	−286	c_{19}	−44755938558	c_{35}	44362575159532	c_{51}	48566054114
c_4	−4617	c_{20}	133891907300	c_{36}	49604735271313	c_{52}	5772587962
c_5	68530	c_{21}	−130069250662	c_{37}	42205820529744	c_{53}	−10681715822
c_6	−112293	c_{22}	11453876074	c_{38}	−81706649969981	c_{54}	6469909883
c_7	220148	c_{23}	−215537550168	c_{39}	−26323825447224	c_{55}	−1224848586
c_8	−5157454	c_{24}	−1178396426815	c_{40}	−75556776520216	c_{56}	−206492641
c_9	21305312	c_{25}	1039103224956	c_{41}	−7607976276074	c_{57}	128463480
c_{10}	−61071486	c_{26}	−1552522849640	c_{42}	75228867400657	c_{58}	−47876576
c_{11}	201511150	c_{27}	7590713636338	c_{43}	−15871634335522	c_{59}	15100000
c_{12}	−418283082	c_{28}	1217758347680	c_{44}	−28710814436688	c_{60}	−2045632
c_{13}	796651448	c_{29}	5350212766744	c_{45}	16898139362482	c_{61}	43776
c_{14}	−1322170362	c_{30}	−6056418085218	c_{46}	3820268662185	c_{62}	−256
c_{15}	−183400046	c_{31}	−28340873761054	c_{47}	−5385634061598		
c_{16}	1215030839	c_{32}	−8777298183560	c_{48}	816867507308		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 62, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $62+4$ terms removed returns order 62 as well, so $\deg q_{\min} = 62 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $62+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 62 that this sequence satisfies — and that family turns out to have exactly one member.

References

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